Week 5: from phases to reactions

Read this slide. Open the Week 5 worksheet, page 1. Keep it beside the video. This lesson replaces the class on **6 November**. We return to the classroom on **13 November**.

Week 4 process	What controls equilibrium?
Bulk sublimation	A separate pure solid can coexist with its gas. While solid remains, $P_A = P_{\mathrm{sub}}(T)$ in this model.
Surface adsorption	Molecules occupy an adsorbent surface. An isotherm gives uptake $q(P_A)$ per mass of adsorbent.

Both calculations also need an **inventory balance**. These conditions answer different questions.

Partial and total pressure at fixed volume

Pause for 4 minutes. Worksheet p. 1, R1. Calculate first. Check your answer on slide 4. If stuck, mark the step and continue.

$T = 300$ K, $V = 1$ L, $P_{\mathrm{sub}} = 10$ kPa. Total volatile amount: $n_A = 0.002$ mol.

$$n_{\mathrm{cap}} = \frac{P_{\mathrm{sub}} V}{RT}, \qquad P_A = \frac{n_{A,\mathrm{gas}} RT}{V}$$

1. Does any solid remain? Calculate P_A.
2. Add 0.002 mol of ideal, nonadsorbing inert gas at the **same T and free volume**. Calculate P_{total}.

Use $R = 8.314462618$ kPa L mol^{-1} K^{-1}. State which pressure changes.

A balance needs an independent equation

Pause for 4 minutes. Worksheet p. 1, R2. Substitute the proposed pressure into both equations. Slide 4 gives the check.

A candidate Langmuir state has $P_A = 10$ kPa, $T = 300$ K, $V_g = 1$ L and $m_s = 0.010$ kg. Given $q_s = 2$ mol/kg, $b = 0.1$ kPa^{-1} and total feed 0.010 mol:

$$n_{\text{gas}} = \frac{P_A V_g}{RT}, \qquad n_{\text{ads}} = m_s \frac{q_s b P_A}{1 + b P_A}$$

Calculate each amount **independently**. Does their sum equal the stated feed?

A number obtained by subtracting gas moles from the feed still needs to satisfy the isotherm.

Two calculations, one physical state

Check R1-R2 on worksheet p. 1. Keep your first attempt and write corrections beside it. No extra Lab run is required for this review.

R1: $n_{\mathrm{cap}} = 0.004009$ mol exceeds the available 0.002 mol. No solid remains. $P_A = 4.989$ kPa. Adding the inert gas gives $P_{\mathrm{total}} = 9.977$ kPa while P_A stays unchanged.

R2: At the proposed 10 kPa, $n_{\mathrm{gas}} = 0.004009$ mol and $n_{\mathrm{ads}} = 0.010000$ mol. Their sum, 0.014009 mol, exceeds the feed.

The checked equilibrium is $P_A = 6.080$ kPa, with $0.002438 + 0.007562 = 0.010000$ mol.

Next: chemical equilibrium also combines an equilibrium relation with conserved amounts.

One worksheet, one Lab, three hours

Read the route on worksheet p. 1. Use the indicated worksheet section at every pause. Open the Lab guide only for the Lab 12 steps.

Required work	Planned time
Video explanations and demonstrations	About 42 min
Worksheet, Lab 12, exit ticket and feedback	128 min
Break	10 min

There is **no required preclass reading or follow-up assignment**. Additional models are optional references.

At a pause limit, record the unresolved step and continue. Bring it to class on **13 November**. The time budget is a guide, not a speed test.

Reaction extent keeps the atoms balanced

Read with worksheet p. 2, A1. Use the same synthetic reaction throughout this lesson. You will complete A1-A3 at slide 8.

$$\mathrm{A_2(g)} \rightleftharpoons 2\mathrm{A(g)}$$

Initially: 1 mol of A_2, no A and no inert gas. A is a formal teaching species.

$$n_{A_2} = 1\ \mathrm{mol} - \xi, \qquad n_A = 2\xi, \qquad 0 \le \xi \le 1\ \mathrm{mol}$$

Define the dimensionless fraction $f = \xi/(1\ \mathrm{mol})$. Then $n_{\mathrm{total}} = (1 + f)$ mol.

The conserved atom amount is $2n_{A_2} + n_A = 2$ mol of A atoms.

Activities make Q dimensionless

Read with worksheet p. 2, A2. The Lab 12 extent calculation assumes one ideal gas phase at fixed temperature and total pressure.

For each gas, $a_i = y_i P/p^\circ$, with $p^\circ = 100$ kPa.

$$Q = \frac{a_A^2}{a_{A_2}} = \frac{y_A^2}{y_{A_2}} \frac{P}{p^\circ}$$

- P is the actual **total pressure**.
- p° defines the standard state and stays fixed in our comparisons.
- Q describes a composition. $K(T)$ describes equilibrium for the reaction and standard states as written.

Use dimensionless P/p°, not a pressure carrying units inside a logarithm.

Your turn: build Q from the mole table

Pause for 10 minutes. Worksheet p. 2, A1-A3. Work through the mole table and quotient. Then continue to slide 9.

1. Write the mole amounts, total moles and mole fractions in terms of f.
2. Substitute the mole fractions into Q.
3. Evaluate Q at $f = 0.50$ and $P/p^\circ = 1$, then at $P/p^\circ = 4$.

Check the atom balance at $f = 0.50$.

For finite $\ln Q$, work inside $0 < f < 1$. The pure-feed and complete-conversion endpoints need limiting interpretations.

Check the quotient and the atom balance

Check A1-A3 on worksheet p. 2. Correct the algebra before using it in the following calculations.

$$y_{A_2} = \frac{1 - f}{1 + f}, \qquad y_A = \frac{2f}{1 + f}$$

$$Q = \frac{4f^2}{1 - f^2}\frac{P}{p^\circ}$$

At $f = 0.50$: $n_{A_2} = 0.50$ mol, $n_A = 1.00$ mol, $y_A = 2/3$ and $y_{A_2} = 1/3$.

$Q = 1.3333$ at $P/p^\circ = 1$ and $Q = 5.3333$ at $P/p^\circ = 4$.

The atom check is $2(0.50) + 1.00 = 2.00$ mol of A atoms. These are specified trial states, not yet equilibrium solutions.

Which way does the reaction proceed?

Pause for 6 minutes. Worksheet p. 2, B1. Calculate the driving force at each trial Q, then predict forward or reverse reaction. Check on slide 14.

At 350 K, the supplied standard reaction Gibbs energy is $\Delta_r G^\circ = -4000$ J/mol of reaction.

$$\Delta_r G = \Delta_r G^\circ + RT \ln Q$$

Evaluate $Q = 1$ and $Q = 10$, using $R = 8.314462618$ J mol^{-1} K^{-1}.

Negative $\Delta_r G$ favors increasing extent. Positive $\Delta_r G$ favors decreasing extent. An interior equilibrium has $\Delta_r G = 0$.

Route 1: K from standard Gibbs energy

Read with worksheet p. 3, B2. Write the substitution now. Complete the numeric comparison at slide 13.

$$\ln K = -\frac{\Delta_r G^\circ}{RT}$$

At 350 K, use $\Delta_r G^\circ = -4.000$ **kJ per mol of the reaction as written**.

Convert kJ to J before using $R = 8.314462618$ J mol^{-1} K^{-1}.

Calculate $\ln K$ first, then $K = \exp(\ln K)$. A standard Gibbs energy supplied at one temperature is a point value.

Route 2: K from enthalpy and entropy

Read with worksheet p. 3, B3. Keep the same reaction, temperature and standard states as Route 1.

At the reference temperature 350 K, the synthetic definitions are:

$$\Delta_r H^\circ = 50000\ \mathrm{J/mol}, \qquad \Delta_r S^\circ = \frac{54000}{350}\ \mathrm{J\,mol^{-1}K^{-1}}$$

$$\Delta_r G^\circ = \Delta_r H^\circ - T\Delta_r S^\circ$$

Then use the same $\ln K = -\Delta_r G^\circ/(RT)$ relation.

An endothermic reaction can have $K > 1$: the entropy term also contributes to the standard Gibbs energy.

Your turn: compare the two K routes

Pause for 10 minutes. Worksheet p. 3, B2-B4. Calculate both routes before opening the worked solutions. Continue to slide 14 for the check.

Route	Calculate
Direct standard Gibbs energy	$\ln K$ and K at 350 K
Standard enthalpy and entropy	$\Delta_r G^\circ$, $\ln K$ and K at 350 K

Explain why agreement is expected for these **mutually consistent synthetic inputs**.

Does agreement between two calculations using those definitions independently validate a real chemical system?

Both routes give the same reference K

Check B1-B4. Mark the energy units and the sign in your worksheet. Retain these values for the remaining cases.

$$\Delta_r G^\circ = 50000 - 350 \left(\frac{54000}{350} \right) = -4000 \text{ J/mol}$$

$$\ln K = 1.3745412, \qquad K = 3.9532626$$

B1: $Q = 1$ gives $\Delta_r G = -4.000$ kJ/mol, so forward reaction is favored. $Q = 10$ gives $\Delta_r G = +2.7007$ kJ/mol, so reverse reaction is favored.

Agreement checks consistent use of the supplied definitions. Real-system prediction still needs valid property data and an appropriate model.

Your turn: solve for equilibrium extent

Pause for 15 minutes. Worksheet p. 3, C1-C3. Solve the reference case by hand before checking slide 16.

At 350 K, $K = 3.9532626$, $P = 100$ kPa and $p^\circ = 100$ kPa:

$$K = \frac{4f^2}{1 - f^2}\frac{P}{p^\circ}$$

1. Rearrange for f^2, then take the physically feasible positive root.
2. Calculate ξ, n_{A_2}, n_A, y_{A_2} and y_A.
3. Recompute Q and the conserved atom amount from your result.

Use $f = \xi/(1\ \mathrm{mol})$, so the mole and dimensionless quantities remain distinct.

The reference equilibrium satisfies Q=K

Check C1-C3 on worksheet p. 3. Use unrounded values for a tight numerical check.

$$f = \sqrt{\frac{K}{K + 4P/p^\circ}} = 0.7050261, \qquad \xi = 0.7050261\ \text{mol}$$

Quantity	Reference equilibrium
n_{A_2} / mol	0.2949739
n_A / mol	1.4100521
y_{A_2}, y_A	0.1730026, 0.8269974

$Q = 3.9532626 = K$ and $2n_{A_2} + n_A = 2.0000000$ mol of A atoms.

Your turn: compress at the same temperature

Pause for 8 minutes. Worksheet p. 4, C4. Predict the direction first, then calculate. Check on slide 18.

Keep 350 K, the reaction, initial inventory and $p^\circ = 100$ kPa. Raise the actual total pressure from 100 to **400 kPa**.

- Does standard K change?
- Which way should equilibrium extent move for $\mathrm{A}_2 \rightleftharpoons 2\mathrm{A}$?
- Calculate f, ξ and the equilibrium mole fractions with $P/p^\circ = 4$.

This is a fixed-T, P reaction calculation. Do not replace it with the fixed-volume inert-gas argument from Week 4.

Pressure changes composition at fixed K

Check C4 on worksheet p. 4. Compare the same quantities in both cases.

At 350 K	100 kPa	400 kPa
Standard K	3.953263	3.953263
ξ / mol	0.705026	0.445114
y_A	0.826997	0.616026
y_{A_2}	0.173003	0.383974

At a fixed trial composition, raising pressure increases Q because $\Delta\nu = +1$. Equilibrium shifts toward smaller extent until $Q = K$ again.

Lab 12: prepare a controlled comparison

Pause for 5 minutes. Worksheet p. 4, D1; Lab guide p. 1. Open Lab 12, select **Reset synthetic example**, and inspect the listed controls. Then watch slide 20.

Set $T = T_{\mathrm{ref}} = 350$ K, $P = p^{\circ} = 100$ kPa and inert gas $= 0$ mol.

Change the default $\Delta_r C_p^{\circ}$ from **30 to 0 J mol^{-1} K^{-1}**. Keep the supplied $\Delta_r G^{\circ}$, $\Delta_r H^{\circ}$, $\Delta_r S^{\circ}$ and reference $\ln K$.

For **K source for extent and driving force**, choose **Direct $\Delta_r G^{\circ}$ at its supplied T**. Verify the reaction table starts with 1 mol A_2 and 0 mol A.

Lab 12 demonstration: the reference state

Watch the actual Lab demonstration. Follow guide p. 1. Observe the result before beginning D2-D4 on slide 21.

1. Click **Compare K methods** and inspect the reference values.
2. Read **From K to equilibrium composition**: extent, equilibrium moles, activities and atom balance.
3. Select $\Delta H^\circ / \Delta S^\circ$ · **constant values**, recalculate and compare the reference result.
4. Return to **Direct** $\Delta_r G^\circ$, recalculate and export **12A**.

The separate **Observed activities** panel uses supplied trial values $y_{A_2} = 0.2, y_A = 0.8$. It does not automatically show the extent solver's equilibrium composition.

Your turn: reproduce the pressure comparison

Pause for 22 minutes. Worksheet p. 4, D2-D4; guide p. 1. Run the cases yourself. Slide 22 provides the pressure demonstration and check.

- **12A:** 350 K, 100 kPa. Compare direct Gibbs with the enthalpy/entropy route at this temperature.
- **12B:** keep 350 K and change only pressure to 400 kPa. Use direct Gibbs for the saved case.
- Recalculate after every change. Record K, ξ, mole fractions and one atom/Q check. Export both studies.

Identify both axes and the minimum in the reaction Gibbs-energy graph. If blocked, record the control or message and continue at the time limit.

What the Lab results establish

Watch the 400 kPa check and compare D2-D4. Correct your own records. The worked solutions contain the same reference numbers.

The graph plots reaction extent ξ in mol against relative total Gibbs energy $[G(\xi) - G(0)]/(RT)$ in **mol**.

The selected equilibrium is the minimum within this admitted **single neutral ideal-gas reaction model**. At the solution, $\ln Q - \ln K \approx 0$ and the atom balance closes.

At 400 kPa, K remains 3.953263 while ξ decreases to 0.445114 mol. A successful calculation checks this model; it does not establish real-gas accuracy or rule out other phases.

Ten-minute break

Pause for 10 minutes. Save your worksheet and the two exported studies. Resume at slide 24.

Your first comparison changed **pressure at fixed temperature**.

After the break, change **temperature at fixed pressure** and ask why K now changes.

You do not need to complete optional reading during the break.

Temperature changes the equilibrium constant

Read with worksheet p. 5, E1. Keep $P = p^\circ = 100$ kPa for the next case.

The direct value $\Delta_r G^\circ = -4.000$ kJ/mol was supplied at **350 K**. It does not describe every temperature.

For this lesson, define a constant reaction enthalpy of $+50.000$ kJ/mol over 350-400 K, equivalent here to $\Delta_r C_p^\circ = 0$.

The positive enthalpy predicts that increasing temperature increases K for the reaction as written.

Route 3: K at a new temperature

Read with worksheet p. 5, E1-E2. Use the reference constant calculated earlier and the stated constant-enthalpy assumption.

$$\ln\left[\frac{K(T)}{K(350)}\right] = \frac{\Delta_r H^\circ}{R}\left(\frac{1}{350} - \frac{1}{T}\right)$$

For $T = 400$ K, $\Delta_r H^\circ = 50000$ J/mol and $\ln K(350) = 1.3745412$.

First calculate the increment in $\ln K$, then add it to $\ln K(350)$. Exponentiate once at the end.

This temperature law needs a valid approximation over the interval, consistent reference data and no intervening phase transition.

Your turn: heat from 350 K to 400 K

Pause for 15 minutes. Worksheet p. 5, E1-E3. Calculate before watching slide 27.

1. Calculate $\ln K(400)$ and $K(400)$ using van't Hoff.
2. Cross-check with $\Delta_r G^\circ(400) = 50000 - 400(54000/350)$ J/mol under the same $\Delta_r C_p^\circ = 0$ assumption.
3. At $P/p^\circ = 1$, calculate f, equilibrium mole fractions and the direction of change relative to 350 K.

Keep pressure, initial inventory and standard states fixed during this temperature comparison.

The endothermic reaction dissociates further

Check E1-E3 on worksheet p. 5. Compare logarithms before comparing rounded K values.

$$\ln K(400) = 1.3745412 + 2.1477206 = 3.5222618$$

$$K(400) = 33.8609295, \qquad \Delta_r G^\circ(400) = -11.714286\ \mathrm{kJ/mol}$$

At 100 kPa: $\xi = 0.945701$ mol, $y_A = 0.972093$ and $y_{A_2} = 0.027907$.

The initial atom inventory stays fixed while temperature changes the standard equilibrium constant and the equilibrium composition.

A 350 K Gibbs value is not a temperature law

Read with worksheet p. 5, F1 and guide p. 2. Identify the applicable K source before changing the evaluation temperature.

At 400 K, choose **van ’t Hoff · constant** ΔH° for the extent solver. Retain $T_{\mathrm{ref}} = 350$ K, the reference $\ln K$ and $\Delta_r C_p^\circ = 0$.

Direct and formation Gibbs-energy inputs supplied at 350 K become **unavailable at 400 K**. Changing their temperature labels would claim new property data; it would not calculate them.

Custom species, nonzero heat capacities and more detailed temperature models are **optional references**, outside this lesson’s required work.

Your turn: save and reopen the temperature case

Pause for 19 minutes. Worksheet p. 5, F1-F3; guide p. 2. Run the new case and verify a restored file before the demonstration on slide 30.

- **12C:** select constant-enthalpy van't Hoff, set $T = 400$ K and $P = 100$ kPa, then recalculate. Keep all reference values fixed.
- Record the applicable K, extent, composition and atom/Q check. Export the complete study.
- Reopen your **12A** study with **Restore calculator study JSON**. Verify the restored settings and freshly recalculated result.

If a value is unavailable, record the reason. Do not replace it with zero. Record unresolved steps and continue when the pause budget ends.

Three cases separate pressure and temperature

Watch the 400 K calculation and JSON restore. Check F1-F3 and your three saved records. Worksheet p. 6 lists what to retain.

Case	T / K	P / kPa	K	ξ / mol
12A	350	100	3.953263	0.705026
12B	350	400	3.953263	0.445114
12C	400	100	33.860929	0.945701

12A to 12B isolates pressure. 12A to 12C isolates temperature under the specified constant-enthalpy model.

A restored study must reproduce its settings and fresh calculation. Exporting a JSON file does not submit it to the instructor.

Bring one unresolved question to our next class

Read worksheet p. 6, G. Prepare one concise explanation and one unresolved step, if any. We will discuss these in class on **13 November**.

- K uses the reaction, temperature and standard-state convention.
- Q uses a particular composition and pressure. At equilibrium, $Q = K$.
- Extent converts the initial inventory into a composition while conserving atoms.
- A pressure change can alter composition without changing K. A temperature change generally changes K.

Keep one worksheet and three exported studies. Optional material is not a requirement for completion.

Optional definitions: IUPAC standard equilibrium constant · IUPAC equilibrium constant.

Exit ticket, feedback and saved work

Pause for 10 minutes. Worksheet p. 6, G. Open the Session 5 exit ticket and feedback form. Submit once and wait for **Response received**.

In your explanation, distinguish K from Q and explain the pressure comparison. Give one numerical result or independent check, with settings and units.

In feedback, report actual study minutes, whether the pace was suitable and the exact step you want explained in class. English or Thai is welcome in the form.

The form is anonymous and does not identify individual attendance. Save your worksheet, corrections and JSON files. Your required Week 5 work ends here.

Open Session 5 form